Course: APT 3020 – Knowledge-Based Systems

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Lab 2: Reasoning and Inferencing in Expert Systems

Explanation Report: Academic Advisor System

1. System Overview

The Academic Advisor System is a Knowledge-Based System (KBS) designed to evaluate student performance metrics and output academic standing recommendations. The system automates complex administrative evaluations by representing institutional regulations as logical rules.

2. Knowledge Representation

The domain knowledge is represented using a dictionary-based structure (exported as knowledge_base.json). This ensures that the logical conditions (e.g., GPA thresholds, attendance requirements, fee statuses) are kept strictly separate from the execution code.

3. Reasoning Methodology: Forward Chaining

The primary reasoning engine utilizes forward chaining (data-driven inferencing).

1. **Fact Extraction:** The system first ingests raw student data (e.g., GPA: 3.8, Attendance: 90%) and translates these into a set of verified system facts (e.g., GPA > 3.5, Attendance > 80%).
2. **Rule Matching:** The engine iterates through the Knowledge Base, checking if the required conditions for each rule are a subset of the student's verified facts.
3. **Execution:** If a match is found, the rule fires and generates a conclusion. Because it evaluates all rules against the facts, it can successfully generate multiple concurrent conclusions (e.g., a student being eligible for both a Scholarship and the Dean's List).

4. Explanation Facility

To ensure the system is transparent, an explanation facility was integrated into the inference engine. When a rule fires, the system concatenates the specific subset of facts that triggered it.

- *Example Trace:* "✓ Eligible for Scholarship because: No Disciplinary Cases, Attendance > 80%, GPA > 3.5" This allows academic staff to audit exactly *why* the system made a specific recommendation.

5. Bonus Implementation: Backward Chaining

To contrast the data-driven approach, a backward chaining (goal-driven) module was also implemented.

- **Process:** Instead of starting with data, the engine starts with a goal (e.g., "Eligible for Graduation"). It retrieves the rule for that goal and checks the system facts to see if the prerequisites are met.
- **Comparison:** While Forward Chaining was highly efficient for generating a complete student profile, Backward Chaining proved superior for targeted queries, as it only evaluates the specific facts necessary for the requested goal, ignoring irrelevant data.